

Christopher Jacobs

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EDUCATION

CLEMSON UNIVERSITY

PH.D. IN COMPUTER SCIENCE

Present | Charleston, SC

GPA: 4.0 / 4.0

CLEMSON UNIVERSITY

MS IN DIGITAL PRODUCTION ARTS

May 2024 | Charleston, SC

GPA: 3.96 / 4.0

FRANCIS MARION UNIVERSITY

BS IN COMPUTER SCIENCE

May 2022 | Florence, SC

Magna Cum Laude | University Honors

GPA: 3.93 / 4.0

LINKS

Github:// [christopher-jacobs](#)

LinkedIn:// [christopher-eugene-jacobs](#)

SKILLS

Languages

Python • C++ • Java • SQL

Frameworks

PyTorch • NumPy • OpenCV • Matplotlib

• PyQt 5 • PySide 6 • OpenGL • TaiChi

Tools

Unreal Engine 4/5 • Autodesk Maya •

Houdini • Nuke • Perforce

Technical Skills

Version Control (Git) • Linux • Bash

Scripting

Research Areas

Neural Inverse Rendering • Computer

Vision • Material Acquisition • Object

Reconstruction

COURSEWORK

Deep Learning in Computer Vision

Physically Based Modeling & Animation

(*Research Asst.*)

Computer Graphics

Technical Art & Direction

Software Engineering

(*Teaching Asst.*)

Physically Based Visual Effects

Cinematography

3D Modeling and Animation

Studio Production

EXPERIENCE

TEACHING ASSISTANT | CLEMSON UNIVERSITY

Jan 2026 – Present | Charleston, SC

- Collaborate with multiple professors to assist over 100 students in software engineering courses through one-on-one office hours and curated assignments.
- Tutored students in software engineering principles including Agile software development, version control through Git, and how to utilize REST APIs.

RESEARCH ASSISTANT | CLEMSON UNIVERSITY

Jan 2024 – Dec 2025 | Charleston, SC

- Engineered a real-time middleware system between Project Chrono (Python) and Unreal Engine (C++) to enable high-fidelity physical simulations for prototyping and visualization.
- Designed a hybrid physically-informed neural network (PINN) framework that could accelerate vehicle simulation in Unreal Engine by reducing computational overhead while maintaining physical accuracy.

ASSISTANT MANAGER | AMERICAN EAGLE OUTFITTERS

Jan 2021 – May 2022 | Florence, SC

- Directed and optimized selling team and merchandise team performance to increase store revenue (+\$5,000 from 2021 LY) and conversion rates (+2.5% from 2021 LY).
- Train and onboarded new associates on store procedures, product knowledge, and selling models while maintaining a welcoming and inclusive environment for both guests and brand ambassadors.

PROJECTS

PHYSICALLY BASED RAY MARCHER FOR VISUAL EFFECTS

Implemented a physically accurate ray marcher using **C++** for volumetric effects including spline wisps, noise clouds, and terrain synthesis, all of which mimic film-quality VFX pipelines.

GEOMETRY INFERENCE USING U-NET FEATURE NETWORK

Using **TEMPEH's** U-Net volumetric feature network and **TaiChi**, implemented a multi-threaded neural inference **Python** program to visualize sensor views, data, and emulate COLMAP's SfM output

AWARDS

2025 Best Presentation Award at MIG

2024 Digital Production Arts Award

2022 Pee Dee Idea Challenge Finalist (4th/100)

2022 Upsilon Pi Epsilon Inductee

2022 University Honors

PUBLICATIONS

- [1] E. Port, C. Jacobs, and J. T. Kider Jr. Understanding player dynamics in battle royale environments: A data-driven analysis using the caldera dataset. In *Proceedings of the 2025 18th ACM SIGGRAPH Conference on Motion, Interaction, and Games*, pages 1–11, 2025.